BeYond COVID - FAIR and open data sharing in support of European preparedness for COVID-19 and other infectious diseases

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ELIXIR All Hands 2022, 7-10 June, Amsterdam, Netherlands

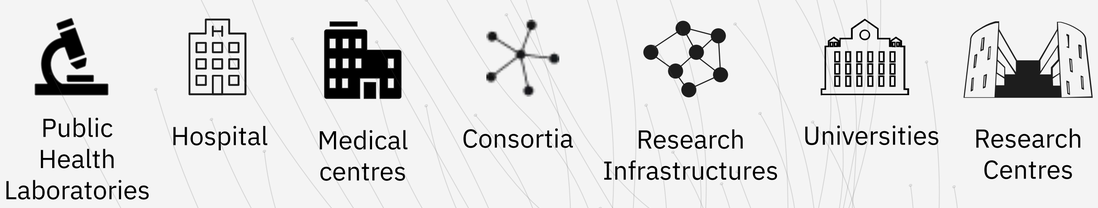
BeYond-COVID (BY-COVID) aims to provide comprehensive open data on SARS-CoV-2, and other infectious diseases across scientific, medical, public health and policy domains. The project has a strong emphasis on mobilising raw viral sequences, helping to identify and monitor the spread of SARS-CoV-2 variants. It further accelerates access to and linking of data and metadata on SARS-CoV-2 and COVID-19 (e.g., public health data, data on patient outcomes), enables federated data analysis to conform with data protection regulations, and harmonisation and management of meta-data and sample-identifiers, as well as long-term cataloguing to ensure interoperability of national and global efforts. BY-COVID builds on the One-Health approach building on the latest technological advances, exploiting and contributing to the European Open Science Cloud capabilities for data access and federation as well as relevant standards and policies for managing, sharing and reusing research data.

The BeYond-COVID project makes **COVID-19 data** accessible to everyone, scientists in laboratories, medical staff in hospitals or government officials.

Going beyond SARS-CoV-2 data, the project will provide a framework for making data from other infectious diseases **open and accessible** to everyone, and build a bridge between public health, clinical sciences, social sciences, and molecular sciences.

The challenge

The world has generated vast amounts of data in response to the COVID-19 pandemic. This data comes from many different sources, and identifying, connecting and integrating these for effective analysis remains a challenge.



Data from SARS-CoV-2, other pathogens, their hosts, and the associated diseases come from many different sources.

The project in numbers

12 M euros

53 partners

36 months

19 countries

What are we doing

1. Mobilising data
2. Connecting data
3. Standardising data
4. Exposing and analysing data

COVID-19 Data Portal

The outcomes

Make SARS-CoV-2 and other infectious disease data easier to access, aggregate and analyse to:

- Enable **scientists in academia and industry** to respond faster to new SARS-CoV-2 strains or other pathogens causing infectious diseases.
- Help **policy-makers** assess the impact of COVID-19 and take appropriate measures to protect people.
- Facilitate **open innovation** in academia and industry across disciplines.

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BY-COVID

This project has received funding from the European Union's Horizon Europe research and innovation programme

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